



Section 1: Case Study – Toll Road Operator

Relates to Questions 1-18

120 Marks available – Time available is 120 minutes – This section has a weighting of 55%

INTRODUCTION

1. You work for a large pension fund called GlobalMutual ('GM'). Currently, your team is preparing an indicative bid to acquire an operational toll road (also known as a turnpike) called Metro Connect ('MC'). You have been asked to build a quarterly "three way" financial model¹ that ultimately can be used to value this business. To assist, an assumptions sheet has been provided. Also provided are partially filled out calculation sheets from GM's templates, showing the layout of the Income Statement and Balance Sheet.
2. The forecast acquisition date is 31 December 2013. GM's internal valuation methodology for this asset is to prepare a quarterly DCF model based on cash flows to/from equity. The projection period should run for 20 years from 1 January 2014 to 31 December 2033. At this time, the rights to charge tolls (the "tolling concession") ends and the asset will be handed back to the local Government. This date is referred to as the "concession end date".
3. GM advises that a discount rate of 15% for Metro Connect should be used in the valuation, and that their convention is to use Excel's XNPV function for valuations. The 15% discount rate is effectively "post-tax" (see the 'Taxation' section below for further detail).
4. The model you develop will be reviewed in detail by another modelling team in your organisation. In order to pass their detailed review it is imperative that you adhere closely to GM's corporate modelling standards as displayed in the formatting legend (included in the model) and the accompanying Income Statement and Balance Sheet layouts. Particular emphasis will be paid to:
 - a. Cell and font colour conventions, units labelling, and row sums;
 - b. Formulae consistency across rows; and
 - c. General layout and clarity, including different sections of the model being easily identifiable
5. You will be allocated a 'formatting mark' by the judges between 0% and 100% based on how closely your model adheres to the GM corporate formatting standards. Your aggregate mark from the multiple choice questions will then be multiplied by the formatting mark to arrive at your final score for this question. *(Note from Judges: We will not be too pedantic here. 100% should be easily achievable if you keep the model neat and follow the guidelines.)*

To assist with formatting, a legend is provided in the workbook with most formats you will wish to use. A Custom Style for each format in the legend has also been added to the workbook.

¹ A "three way" financial model includes an income statement, balance sheet and cash flow statement.



THE BRIEF

Revenues

6. MC generates 100% of its revenue from tolls paid by drivers. These tolls are received either in cash or through its electronic tag system. Different tolls apply for cash trips and tag trips. Toll prices are set by vehicle class, and are derived from the 1 January 2014 starting theoretical tolls provided in the assumptions sheet.
7. The theoretical toll (price) will escalate every 1 July by 2.5%. The charged toll in any quarter will then be:
 - a. For Tag trips, the theoretical toll in that quarter rounded DOWN to the nearest cent.
 - b. For Cash trips, the Tag trip price rounded UP to the nearest ten cents, plus an additional \$0.30 premium (this additional premium does not escalate with time).
8. To assist you in determining the traffic expected, you have received a forecast of Average Daily Traffic ('ADT') for each vehicle class for Q1 2014 as well as the escalation profile for increasing the ADT forecast each quarter. These are included within the assumptions worksheet. Apply escalation on the first day of each quarter. Do not round fractional trips.
9. It is assumed that for every vehicle class, 70% of all trips will be Tag Trips. The remainder will be Cash trips. This proportion is not expected to change over time.
10. For some tag trips, it is expected that tolls charged will not be collected. This lost revenue is known as 'Revenue Leakage' and is forecast to apply to 2% of all tag trips for each vehicle class. Count this as a negative revenue, not as an operating cost. Total Revenue should be reported clearly in the model net of this leakage.

Expenses

11. MC incurs variable operating costs of \$0.80 (real dollars) per tag trip, and \$1.05 (real dollars) per cash trip.
12. MC incurs fixed operating costs. These are detailed in the assumptions worksheet in real dollar terms and are per quarter amounts.

Escalation

13. The following escalation conventions and units labelling and methodology of real and nominal amounts should be adopted:
 - a. [\$r] represent the 'real' dollars as specified in Q1 2014 and will need to be escalated in line with a Consumer Price Index ('CPI').
 - b. [\$n] represent nominal dollars. These are dollars of the day and have already been escalated to the face value that will be transacted on the day.
 - c. Assume an effective (compounded) CPI increase of 2.8% per annum. No escalation should be applied to quarter 1. Escalation should be applied every quarter start date beginning 1 April 2014 on a 90/360 basis. For clarity, an escalated price each 1 January (except 2014) should be exactly 102.8% of the previous 1 January price.

Working Capital

14. Metro Connect's debtors are 10% of total quarterly Tag trip revenue (net of leakage). Its creditors are 10% of total quarterly operating costs.

Balance Sheet

15. The projected opening balance sheet as at 31 December 2013 along with the tax and accounting depreciation schedules for the opening assets are provided in the assumptions worksheet.



Capital Expenditure

16. Cost, timing and useful lives for forecast Capital Expenditure are detailed in the assumptions worksheet. There will be no capital expenditure in the final quarter of the tolling period, so your model logic does not need to attempt to cater for this situation. Calculate the accounting and tax depreciation for capital expenditure assuming purchases occur at the end of the quarter and depreciation should immediately commence on day 1 of the following quarter.
- Accounting depreciation uses the straight-line basis. The useful life for each asset class is provided in the assumptions worksheet.
 - Taxation depreciation uses the Double Declining Balance basis. Continue to depreciate assets at the end of their useful life rather than writing them off². To obtain depreciation for a quarter (D_Q), you should use the following approach:
$$D_Q = 200\% * (WDV_{opening}) / (\text{Useful life in years} * 4).$$
 - At the concession end date, any remaining WDV³ can be immediately fully depreciated for both tax and accounting purposes.
17. All capital expenditure will be 100% funded by cash.

Debt and Interest

18. Assume that all interest relevant to debt is calculated on a 90/360 day basis on the opening balance at the start of the period and is paid on the last day of the quarter. All interest rates given are simple annual rates (not effective-annual compounding rates).
19. MC has an existing senior debt facility of \$120 million on its balance sheet. Interest is payable on the last day of each quarter based on the quarterly opening balance. The simple annual interest rate will be the London Interbank Offered Rate ('LIBOR') plus a 3% margin. Forecast LIBOR rates have been provided in the assumptions worksheet. Scheduled repayments each quarter equal to \$1m nominal have been agreed. No further drawdowns are forecast for this facility.
20. A cash sweep is triggered for any quarter when the outstanding balance on the senior facility immediately after scheduled repayment exceeds \$60.0m. In a quarter where a cash sweep is triggered, cash will be swept until Quarterly Cash For Distribution ('QCFD') is reduced to \$1.0m or the balance of the senior facility reaches \$60.0m. The amount swept cannot be less than zero.
21. QCFD is defined as
- Opening Cash Balance; plus
 - Cash Revenues less Cash expenses (adjusted for working capital); less
 - The sum of tax paid/refunded, cash-funded capex, interest paid on the debt facility, scheduled repayments on the debt facility, and the cash sweep amount.
22. No interest is earned on cash at bank. Likewise, no interest is charged on any overdraft of the cash account. The cash account can be overdrawn.

² Ignore the fact that the tax depreciation on the existing assets will be cut off.

³ Written Down Value



Distributions

- 23. MC pays a dividend of 100% of net profit after tax ('NPAT') but limited such that the closing cash balance post distributions does not fall below \$1.0m. Dividends are paid after all other cash flows on the last day of each quarter.
- 24. On the concession end date, the local Government will acquire all the outstanding shares in Metro Connect for a price equal to the closing balance sheet Equity balance as at 31 December 2033.

Taxation

- 25. MC is a taxable entity assessed quarterly. MC resides in a jurisdiction that does not allow accrual of tax losses. Instead, any taxable loss (i.e. a 'negative' taxable income) will earn a refund equal to the loss amount times the tax rate.
- 26. The tax rate is 24%. Tax is paid/refunded on a one quarter delay.
- 27. MC has an opening Deferred Tax Liability ('DTL') balance representing timing differences on opening assets. All timing related Deferred Tax items should be reported on the balance sheet as a net balance, either as a DTA or DTL.
- 28. GM is exempt from all tax on dividends and any gain (or loss) on the sale of shares. No further tax modelling needs to be applied to the cash distributions paid from MC to GM prior to performing the valuation.
- 29. There is no taxation impact to MC from acquisition on 31 Dec 2013 or on transfer to Government on 31 Dec 2033.